RIG-ETALE ALGEBRAS IN THE CATEGORY C-PRIME ARE COMPLETED FINITE TYPE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2316-2403, following the statement at 2307-2314.

Editorial note. The single (details omitted) at line 2368 is the standard completion base change along an etale (flat) map, a one-step standard-fact instantiation covered by the card's obligations.

Editorial note. The induction is on the number of generators of I; the reduction to the principal case is the organizing device of the whole proof.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 2368 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Principal ideal case. lemma-approximate-principal, cited at proof line 2320. Supplies the t = 1 base case and the final application to B' over (A, (a_t)).
- Base change and etale-over-complement control. lemma-base-change-rig-etale-homomorphism at proof lines 2328-2329 and lemma-rig-etale at lines 2336-2338. Supplies that B_m = B/(a_t^m) stays rig-etale over the reduced pair and that C_m may be chosen etale over the complement of V(I-bar_m).
- Etale lifting machinery. lemma-fully-faithful-etale-over-complement (2350-2354), morphisms-lemma-etale-permanence (2357), more-morphisms-lemma-check-smoothness-on-infinitesimal-nbhds (2359-2360), etale-theorem-remarkable-equivalence (2362-2366), lemma-lift-approximation (2369-2378). Supplies the compatibilization step C_m/(a_t^{m-1}) -> C_{m-1} and the lift C"_m of C_{m-1} to an algebra over A-bar_m.
- NL comparison and rig-etale transfer. lemma-topologically-finite-type (2383-2385), lemma-NL-is-completion (2392-2395), lemma-equivalent-with-artin (2396-2400). Supplies that B' = lim C_m is topologically finite type, that the NL cohomologies agree, and the uniform annihilation transferring rig-etale-ness from B to B'.

Source reference. Original equation 0AL3: sources/stacks-restricted/pinned-original-source.tex, source lines 312–312. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original equation 0AL4: sources/stacks-restricted/pinned-original-source.tex, source lines 326–326. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 039R: focused-seeds/geometry/S9-restricted-approximate/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GAF: sources/stacks-restricted/pinned-original-source.tex, source lines 560–560. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ALS: sources/stacks-restricted/pinned-original-source.tex, source lines 2281–2281. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GB2: sources/stacks-restricted/pinned-original-source.tex, source lines 2048–2048. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AJU: sources/stacks-restricted/pinned-original-source.tex, source lines 1810–1810. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AKJ: sources/stacks-restricted/pinned-original-source.tex, source lines 2065–2065. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ALT: sources/stacks-restricted/pinned-original-source.tex, source lines 2151–2151. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ALQ: sources/stacks-restricted/pinned-original-source.tex, source lines 1981–1981. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0AJP: sources/stacks-restricted/pinned-original-source.tex, source lines 346–346. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A43: focused-seeds/geometry/S9-restricted-approximate/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02GW: focused-seeds/geometry/S9-restricted-approximate/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 0AL6: sources/stacks-restricted/pinned-original-source.tex, source lines 454–454. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let A be a Noetherian ring. Let $I \subset A$ be an ideal. Let B be an object of (0AL4) which is rig-étale over (A, I). Then there exists a finite type A-algebra C and an isomorphism $B \cong C^\wedge$.*

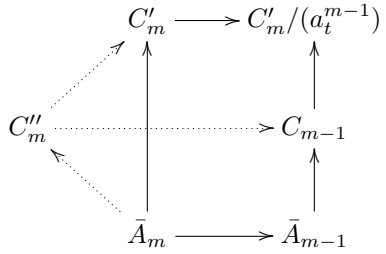
Proof. We prove this lemma by induction on the number of generators of I. Say $I = (a_1, \ldots, a_t)$. If $t = 0$, then $I = 0$ and there is nothing to prove. If $t = 1$, then the lemma follows from Lemma 0ALS. Assume $t > 1$.

For any $m \geq 1$ set $\bar{A}_m = A/(a_t^m)$. Consider the ideal $\bar{I}_m = (\bar{a}_1, \ldots, \bar{a}_{t-1})$ in $\bar{A}_m$. Observe that $V(I\bar{A}_m) = V(\bar{I}_m)$. Let $B_m = B/(a_t^m)$ be the base change of B for the map $(A, I) \to (\bar{A}_m, \bar{I}_m)$, see Remark 0AL6. By Lemma 0GB2 we find that B_m is rig-étale over $(\bar{A}_m, \bar{I}_m)$.

By induction hypothesis (on t) we can find a finite type $\bar{A}_m$-algebra C_m and a map $C_m \to B_m$ which induces an isomorphism $C_m^\wedge \cong B_m$ where the completion is with respect to $\bar{I}_m$. By Lemma 0ALQ we may assume that $\mathrm{Spec}(C_m) \to \mathrm{Spec}(\bar{A}_m)$ is étale over $\mathrm{Spec}(\bar{A}_m) \setminus V(\bar{I}_m)$.

We claim that we may choose $A_m \to C_m \to B_m$ as in the previous paragraph such that moreover there are isomorphisms $C_m/(a_t^{m-1}) \to C_{m-1}$ compatible with the given A-algebra structure and the maps to $B_{m-1} = B_m/(a_t^{m-1})$. Namely, first fix a choice of $A_1 \to C_1 \to B_1$. Suppose we have found $C_{m-1} \to C_{m-2} \to \ldots \to C_1$ with the desired properties. Note that $C_m/(a_t^{m-1})$ is étale over $\mathrm{Spec}(\bar{A}_{m-1}) \setminus V(\bar{I}_{m-1})$. Hence by Lemma 0AKJ there exists an étale extension $C_{m-1} \to C'_{m-1}$ which induces an isomorphism modulo $\bar{I}_{m-1}$ and an $\bar{A}_{m-1}$-algebra map $C_m/(a_t^{m-1}) \to C'_{m-1}$ inducing the isomorphism $B_m/(a_t^{m-1}) \to B_{m-1}$ on completions. Note that $C_m/(a_t^{m-1}) \to C'_{m-1}$ is étale over the complement of $V(\bar{I}_{m-1})$ by Morphisms, Lemma 02GW and over $V(\bar{I}_{m-1})$ induces an isomorphism on completions hence is étale there too (for example by More on Morphisms, Lemma 0A43). Thus

$C_m/(a_t^{m-1}) \to C'_{m-1}$ is étale. By the topological invariance of étale morphisms (Étale Morphisms, Theorem 039R) there exists an étale ring map $C_m \to C'_m$ such that $C_m/(a_t^{m-1}) \to C'_{m-1}$ is isomorphic to $C_m/(a_t^{m-1}) \to C'_m/(a_t^{m-1})$. Observe that the $\bar{I}_m$-adic completion of C'_m is equal to the $\bar{I}_m$-adic completion of C_m, i.e., to B_m (details omitted). We apply Lemma 0ALT to the diagram



to see that there exists a "lift" of C''_m of C_{m-1} to an algebra over $\bar{A}_m$ with all the desired properties.

By construction (C_m) is an object of the category (0AL3) for the principal ideal (a_t). Thus the inverse limit $B' = \lim C_m$ is an (a_t)-adically complete A-algebra such that B'/a_tB' is of finite type over $A/(a_t)$, see Lemma 0AJP. By construction we have $C_m = B'/(a_t^m)$, the I-adic completion of C_m is B_m, and the I-adic completion of B' is isomorphic to B. For each m the complex NL_{C_m/A_m} has finite cohomology modules supported on $V(IC_m) \subset \mathrm{Spec}(C_m)$ by construction. Hence these modules are I-adically complete (as these modules are annihilated by a power of I). Since $NL^\wedge_{B_m/A_m}$ is the I-adic completion of NL_{C_m/A_m}, see Lemma 0GAF, it follows that the map $NL_{C_m/A_m} \to NL_{B_m/A_m}$ induces an isomorphism on cohomologies. Thus because B is rig étale over A it follows that there is a fixed power of I that annihilates the cohomologies of NL_{C_m/A_m} for all m, see Lemma 0AJU. It follows that B' is rig étale over $(A, (a_t))$ by the same lemma. Hence finally, we may apply Lemma 0ALS to B' over $(A, (a_t))$ to finish the proof. □